Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/27/2005
PI Name:	Larson, Melora E. Ph.D.		
Project Title:	Experiments Along Coexistence Near Tricriticality (EXACT (2000))		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Melora.E.Larson@jpl.nasa.gov	**Fax:**	FY 818-393-4878
PI Organization Type:	NASA CENTER	**Phone:**	818-354-8751
Organization Name:	NASA Jet Propulsion Laboratory		
PI Address 1:	Mail stop 79-5		
PI Address 2:	4800 Oak Grove Drive		
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nash, Alfred (Jet Propulsion Laboratory) Mulders, Norbert (University of Delware) Vollmayr-Lee, Ben (Bucknell University)		
Grant/Contract No.:	GSN 4534		
Performance Goal No.:	1.24		
Performance Goal Text:	1.24 To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	EXACT 2000 will measure the second sound velocity as a function of concentration and temperature in mixtures of helium-3 and helium-4 near the tricritical point. From these measurements, the superfluid density will be derived, and the critical behavior of the superfluid density will be found. EXACT 2000 will also use these measurements to measure the shape of the coexistence curve and the Lambda-line as a function of temperature and concentration. The task objective of EXACT 2000 is to make high resolution measurements on mixtures of 3He and 4He to determine the critical exponents at the tricritical point. This task will improve upon existing measurements at the tricritical point by an order of magnitude. These measurements will then be compared to existing, exact theoretical predictions for any possible discrepancy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In 2004, we made steady progress on our research goals. The experimental facility at the University of Delaware was upgraded, and a graduate student took over collecting the second sound data. At JPL, summer undergraduate students contributed to several improvements and upgrades to the phase separation measurement apparatus. We also had 2 papers that we either wrote or contributed to published. Author(s): Dotsenko, V; Mulders, N Title: A really simple cryogenic valve Source: JOURNAL OF LOW TEMPERATURE PHYSICS, 134 (1-2): 443-446 JAN 2004 Author(s): Lammerzahl, C; Ahlers, G; Ashby, N; Barmatz, M; Biermann, PL; Dittus, H; Dohm, V; Duncan, R; Gibble, K; Lipa, J; Lockerbie, N; Mulders, N; Salomon, C Title: Experiments in fundamental physics scheduled and in development for the ISS Source: GENERAL RELATIVITY AND GRAVITATION, 36 (3): 615-649 MAR 2004
Bibliography Type:	Description: (Last Updated: 10/02/2007)
Articles in Peer-reviewed Journals	Dotsenko V, Mulders N. "A really simple cryogenic valve." Journal of Low Temperature Physics, 2004 Jan;134 (1-2): 443-6. http://dx.doi.org/10.1023/B:JOLT.0000012593.10308.2b , Jan-2004
Articles in Peer-reviewed Journals	Lammerzahl C, Ahlers G, Ashby N, Barmatz M, Biermann PL, Dittus H, Dohm V, Duncan R, Gibble K, Lipa J, Lockerbie N, Mulders N, Salomon C. "Experiments in fundamental physics scheduled and in development for the ISS." General Relativity and Gravitation, 2004 Mar;36(3):615-49. http://dx.doi.org/10.1023/B:GERG.0000010734.62571.b4 , Mar-2004